

VertexONE Concepts and Final Designs

Honeywell's next-generation toxic gas monitoring system for semiconductor fabs, an evolution of the award-winning Vertex Edge platform, integrated hardware, embedded interfaces, and cloud-ready HMI software to improve serviceability, ergonomics, and usability.

Designing with AI and Digital Twins for Actionable Insights and Larger system Integration

- **Vertex Insights:** Predictive maintenance and insights based on machine learning from historical fault and alarm data, with the goal of preventing costly downtime.
- **Prediction Score:** Recommended actions with the likelihood of costly events alongside clear indicators that these are AI-generated and should be supplemented with additional data sources.
- **Vertex Assist:** A conversational AI assistant embedded within the device enabling technicians to quickly ask natural language questions for troubleshooting help and maintenance schedules.

VERTEX ONE

Key HMI Differentiators

ALARM/FAULT INTELLIGENCE

Analytics and insights with modern data visualization



Future Release

DIGITIZE MAINTENANCE

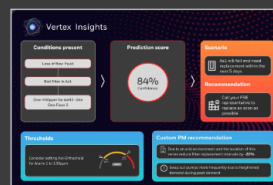
Embedded AI assistant to reduce ticket escalations



Future Release

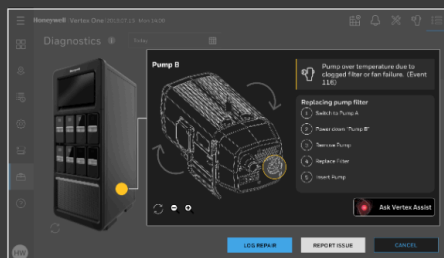
DOWNTIME PREVENTION

Predictive maintenance with AI-generated insights



SIMPLIFY LABOR INTENSIVE TASKS

Advanced diagnostics with guided steps and realistic diagram



FLEXIBILITY (MVO 2)

Remote management of multiple Vertex from various devices.



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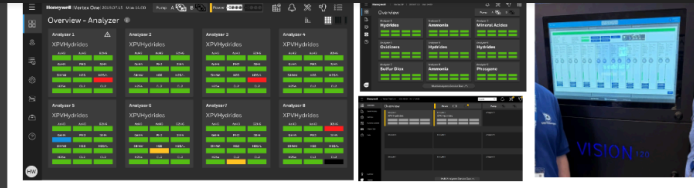
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Matt McGlothlin



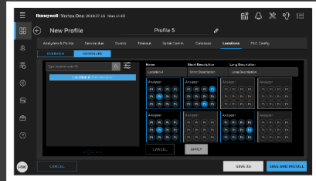
OVERALL EXPERIENCE

Continuous UI improvement (Edge > C > One) strengthens Honeywell usability advantage over major competitors like DoD.



USABILITY

Point mapping system increases productivity and improves recognition/recall of point location configurations.



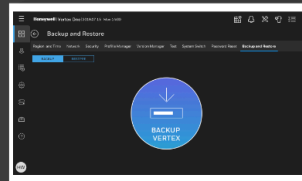
INTEGRATION

TGMS inhibit timer enables FSE to prevent automatic faults



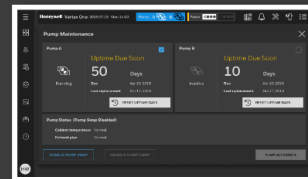
DATA INTEGRITY

Backup and restore prevents data loss and reduces downtime due to unforeseen conflicts.

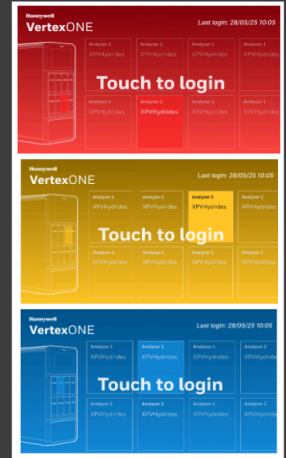


SAFETY

Disabling auto pump swap prevents potential safety issue when working on backup pump.



status.



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REMOTE MANAGEMENT (MVO2)

FEATURES

Multiple Vertex Overview

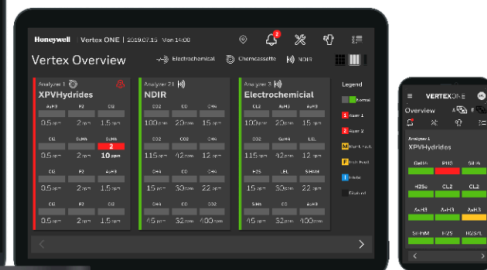
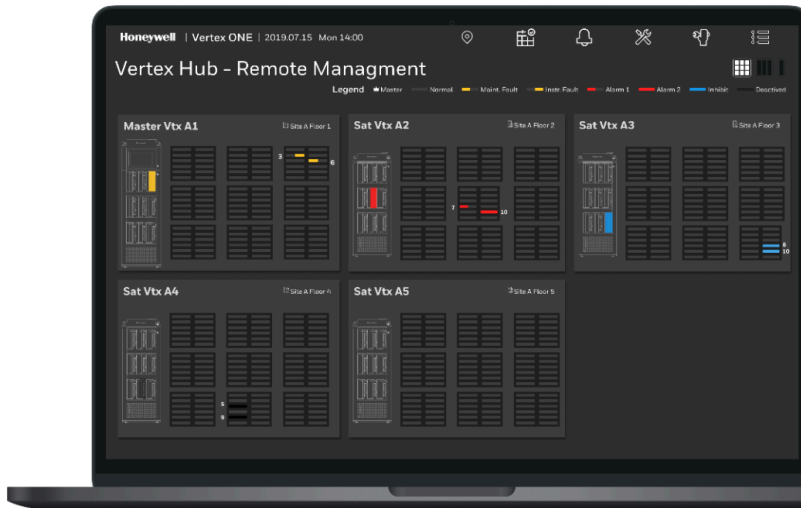
See status of all analyzers/points on all Vertex in one screen including alarms, faults and inhibits.

Drill-down Functionality

Select a Vertex and drill down into the analyzers/points for that Vertex in detail.

Responsive User Interface

View and manage each vertex from table or phone without losing important information.





Power Status

Header iconography allows user to monitor pump and power status in one glance. Selecting either graphic enables user to manage both.

Legend

New users unfamiliar with the UI can touch info icon for legend that defines point color and icons.

Layout Switcher

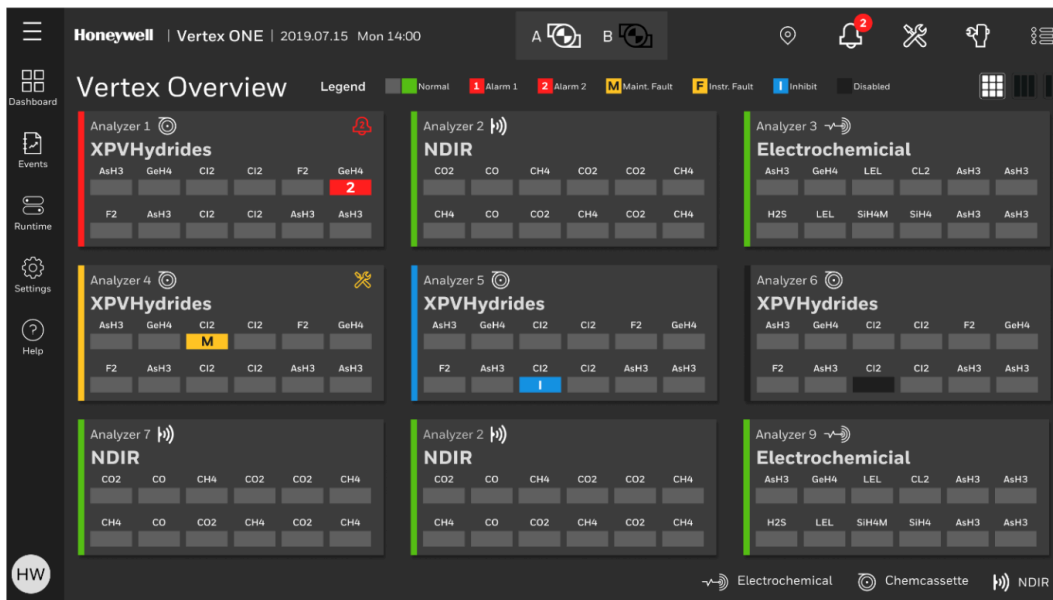
Switching between grid, column and single analyzer view gives users more flexibility and options for viewing analyzer status

Icon-assisted Points

Color blind users are able to distinguish between red and green easily with the help of icons



SENSOR (MVO2)



FEATURES

Iconography

NDIR, Chemcassette and ECC icons signify to user which sensor analyzer contains

Legend

Bottom legend displays definitions for each type of sensor when selecting.

Sensor management

Bump and calibration process for ECC and NDIR sensors similar to Midas M.



FEATURES

Search, Sort and Filter

Search for a specific location or filter by alarm, fault, building. Sort by location name or location status

Point mapping

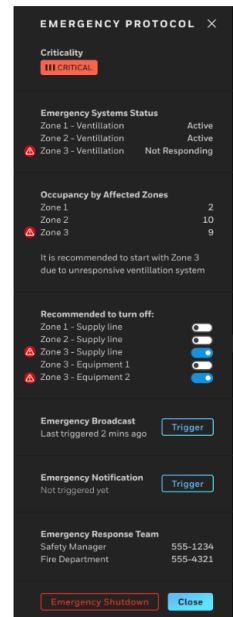
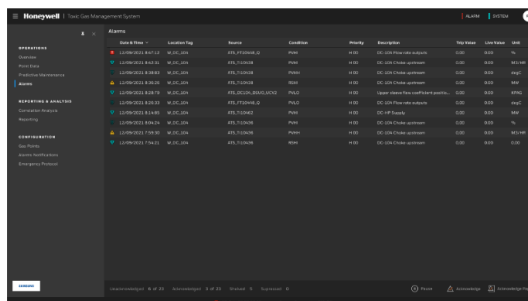
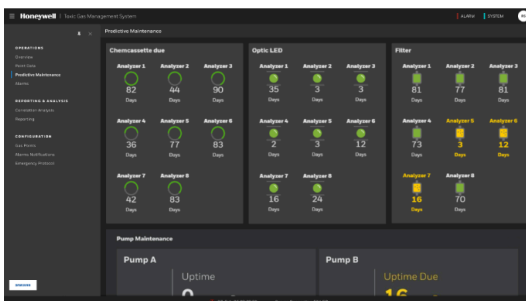
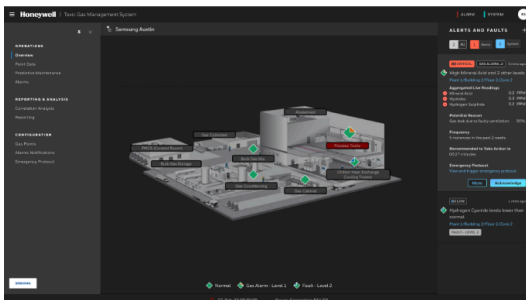
Each locations displays up to six points with Analyzer # and Point #

Gauge Charts

Location detail displays gauge chart with alarm thresholds, concentration and any faults highlighted in yellow.

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TGMS INTEGRATION (WIP)



Concepts, High Fidelity Prototypes and Dev Hand-off

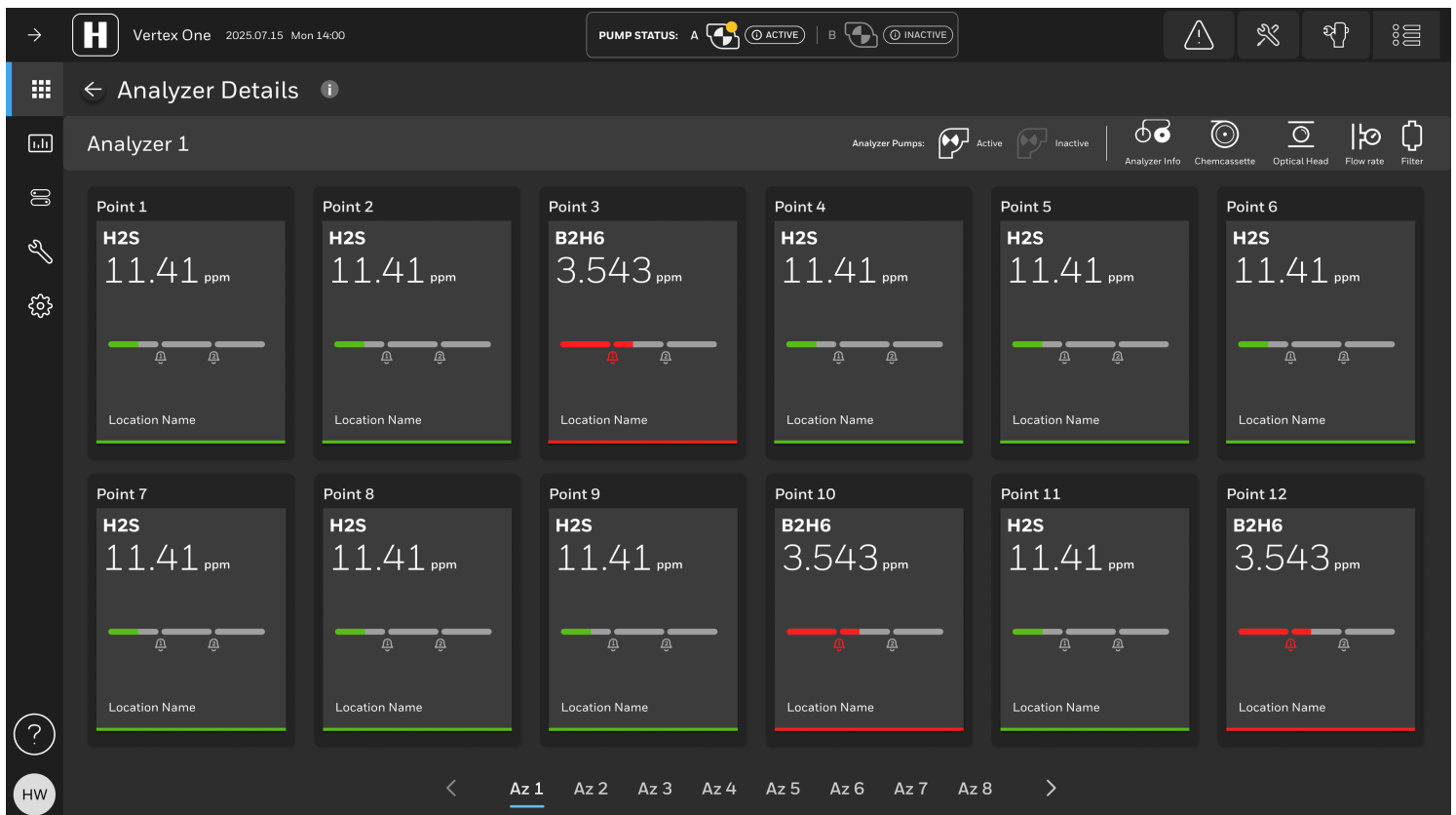
Concepts: Overview Analyzer Details and Point Detail Screens

status in the header is easier to digest and more descriptive of faults since this is one of the differentiators the new Vertex.

Analyzer details show an overview of points with graphical representation of current concentration levels so field technicians can view when a point may be approaching the danger zone.

Chart visualizations and information at the gas detection point level clearly give an indication of when the gas event exceeded thresholds and can be easily exported.





Concept: PLC touch configurator

stadium when purchasing a ticket and mapping that to the number of people attending.

→

H

Vertex One

2025.07.15 Mon 14:00

←

New Profile

i

Analyzer Details

Service Due

Events

Timeouts

Serial Comm

Database

PLC Config

Location

Presets

Fault Relay Mappings

Alarm Relay Mappings

	Card 04	Card 05	Card 06	Card 07
Slot 01	MAINT	AL2	AL2	AL2
Slot 02	BOTH	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL2	AL1	AL1	AL1

	Card 08	Card 09	Card 10	Card 11
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

	Card 12	Card 13	Card 14	Card 15
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

4-15 of 45 items

<

4-15

16-27

28-39

40-45

>

Cancel

Save As

Install

Save

Card

Card

Nothing selected yet

Select a card relay first

→

H

Vertex One

2025.07.15 Mon 14:00

←

New Profile

i

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Slot 04	AL1	AL1	AL1	AL1

4-15 of 45 items

<

4-15

16-27

28-39

40-45

>

Cancel

Save

Save As

Install

Select analyzer points to map

Analyzer 01

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 02

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 03

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 04

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 05

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 06

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 07

01 02 03 04

05 06 07 08

09 10 11 12

Analyzer 08

01 02 03 04

05 06 07 08

09 10 11 12

Select Alarm Type *

☐ Alarm 1

☒ Alarm 2

☐ Both

☐ None

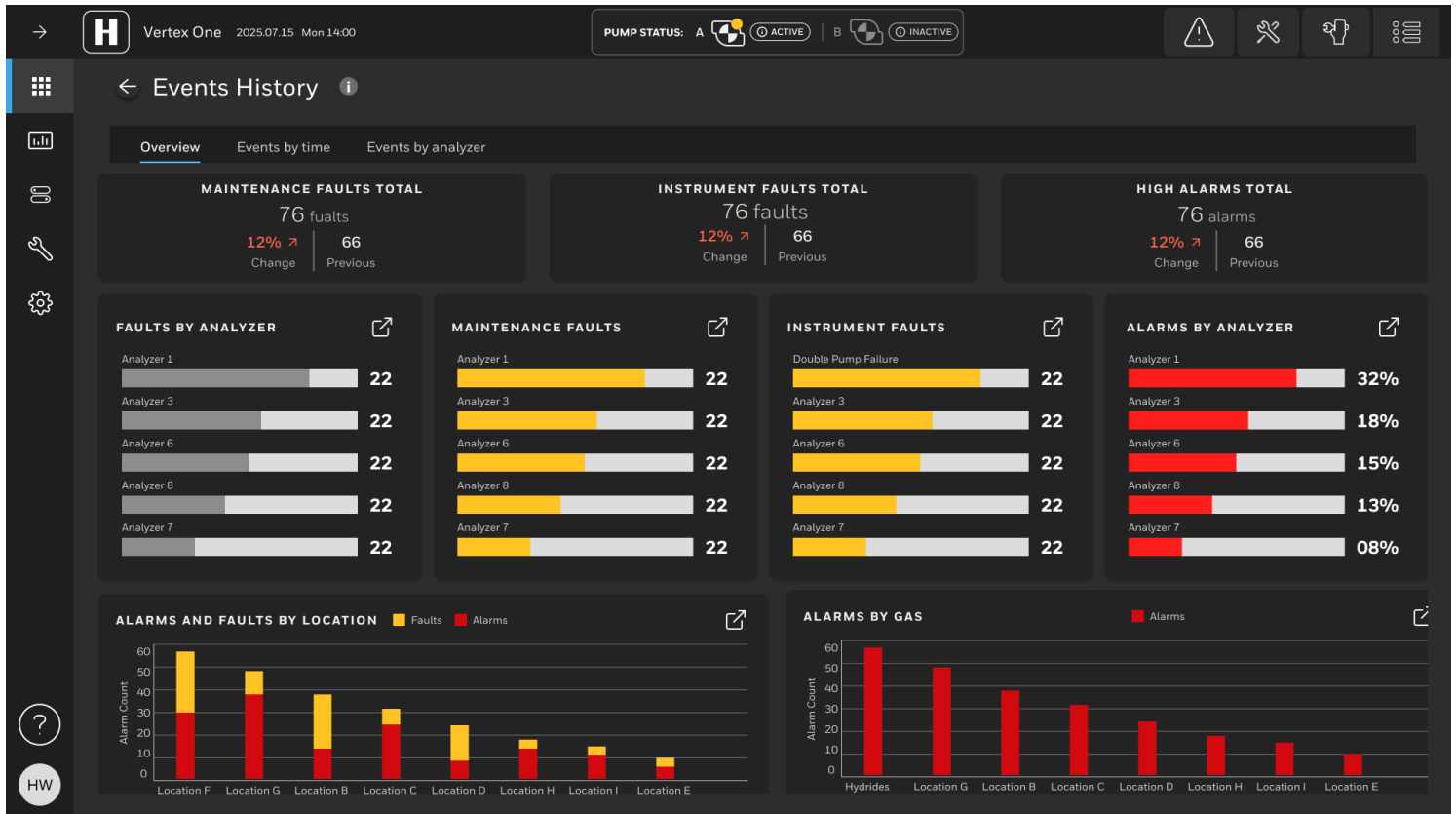
Cancel

Apply

Matt McGlothlin



We provided easy-to-digest visualizations that also helped with the quarterly reporting process for customers. The resulting savings could amount to \$1M+ annually and increase revenue for Honeywell in the high tech space.

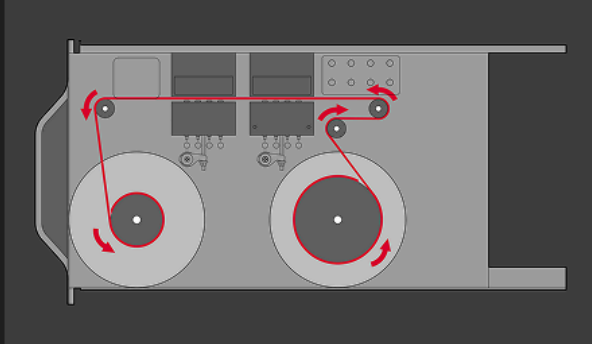


Training and Guided Maintenance

We heard repetitively that documentation was lacking and that training was needed. We solved this by explicitly detailing the steps of complex tasks like replacing Chemcassettes while providing instant and descriptive feedback for any errors. The user can go forward, skip or go back on any step so there is the ability to recover from errors.



Please open each analyzer and replace the chemcassette.
When the all chemcassettes are placed, please click NEXT.



Cancel

Next

- Release analyzer
- Replace Chemcassette
- Uploading data from RFID
- Verify Chemcassette status
- Adjust white, light/dark gray
- Resume monitor or idle

Diagnostics and Recommendations

Ideas circulated around specific recommendations with the rise of artificial intelligence. Internal stakeholders loved this feature including FSEs with many years of experience. The recommendations synthesized from various data sources could be a game changer for Honeywell and allow technicians to spend more time with the customer and less time figuring out how to troubleshoot edge case issues. The digital twin aspect of the diagnostics section received applause from sales, marketing and technicians.

"With the analytics and diagnostics ... This is something we can sell" - Honeywell High Tech President

Analyzer Overview

Analyzer	Chemcassette	Optics	Pump 1	Pump 2	Auto Flow Rate
Analyzer 01	Normal	Normal	Normal	Normal	Normal
Analyzer 02	Service Due Soon	Optics Block Dirt...	Pump1 current o...	Pump2 current ov...	Loss of Flow
Analyzer 03	Service Due Soon	Optics Block Dirt...	Pump1 current o...	Pump2 current ov...	Loss of Flow
Analyzer 04	Normal	Normal	Normal	Normal	Normal
Analyzer 05	Normal	Normal	Normal	Normal	Normal
Analyzer 06	Normal	Normal	Normal	Normal	Normal
Analyzer 07	Normal	Normal	Normal	Normal	Normal
Analyzer 08	Normal	Normal	Normal	Normal	Normal

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Diagnostics

HMI Communications

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Communications

Profile ID

Date Created	08.26.2025
Date Modified	08.31.2025
AZ configurations	8
Event Notifications	Enabled
Login Idle	Enabled

Recommendations

Best Practice

- Use Chrome browser v93.0.xx or later for remote HMI access.
- HTTPS connections require certificate acceptance
- Do not connect the HMI to the internet
- Router with IPSec or Peer-to-Peer (P2P) networking recommended
- Access control is managed via user roles; elevated permissions are required for configuration and maintenance.

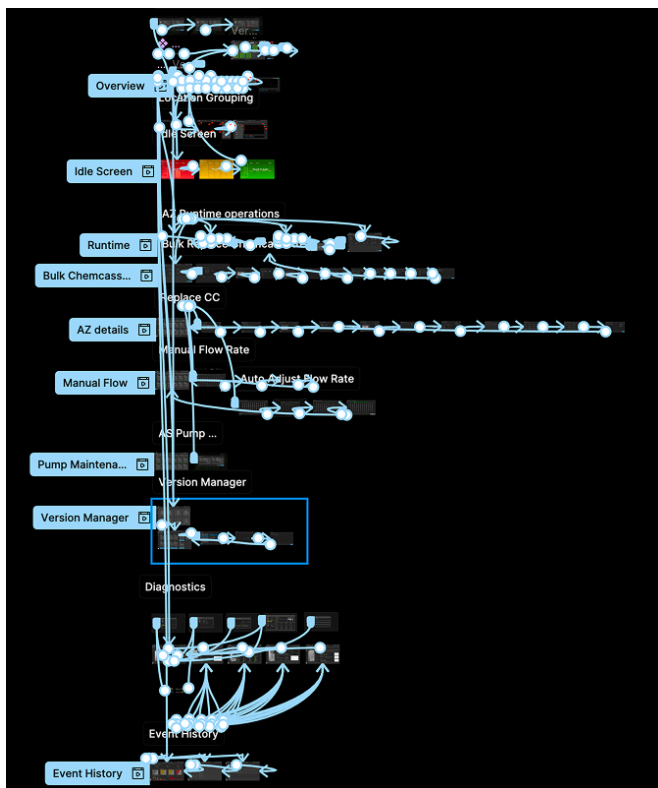
Indicators that suggest replacement

- A frozen screen can mean power surges, improper shutdowns, or file corruption.
- Fault 108 could be failedcomms between analyzer and PC is interrupted.
- Fault 128 could be driver load failure or electronic issue.
- Fault 234 could be Hardware failure.

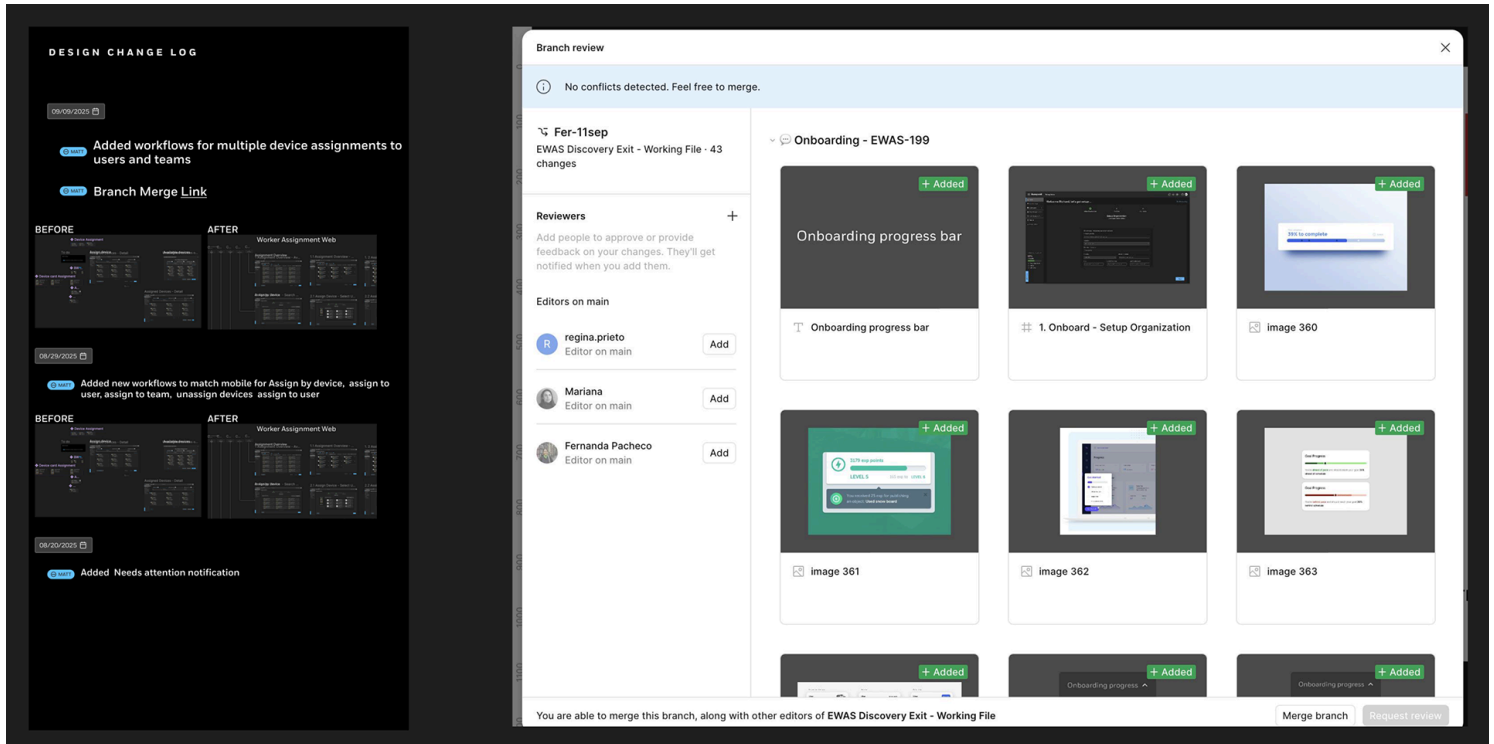


High Fidelity Prototype

The prototype helped to test the final screens with users and internal stakeholders alike. Every screen was linked up to the prototype and we were able to hook it into a tablet that mimicked the actual Vertex machine.



During the dev handoff process it's critical to keep designs updated when there will inevitably be minor tweaks. This is why I implemented a versioning system that included branch and merge functionality similar to GitHub. This native-Figma functionality allowed designers and developers to stay in sync and schedule updates after release in subsequent sprints. Annotating designs is another critical and often overlooked part of the design process. Annotations reduce the amount of back and forth between developers and stakeholders alike.

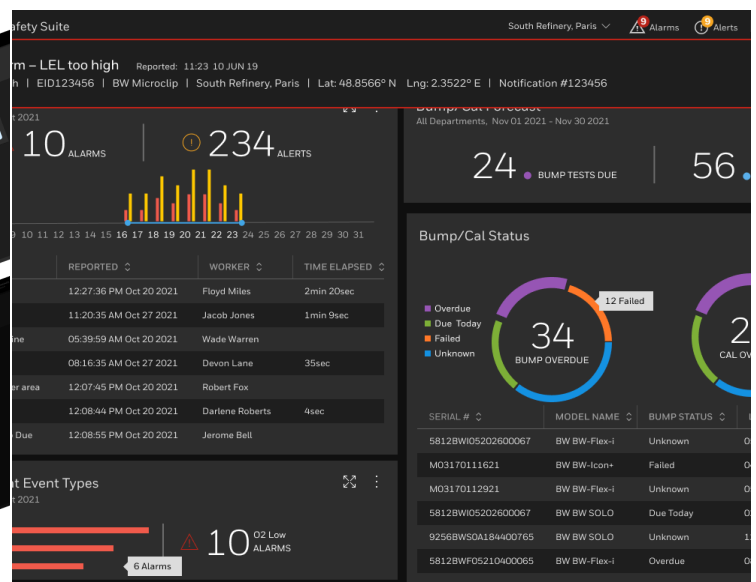
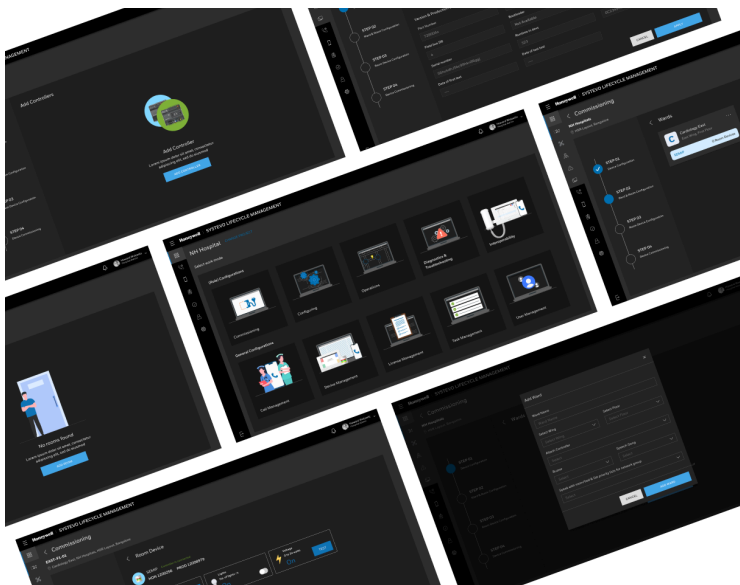


Prototype

The design actions for Vertex One led to an HMI solution recognized for its quality, successfully transforming a fragmented system into a cohesive, mission-critical interface, positioning it as a significant competitive advantage for Honeywell:

Presentations

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